FITNESS ACTIVITY PREDICTICTION

NAME:RIDA YOUSAF
BRANCH:CSE AIML

INSTITUTION:TKM INSTITUTE OF TECHNOLOGY

PROGRAM ABSTRACT:

- Abstract: This project focuses on developing a machine learning model to predict fitness activities (Running, Squats, Walking, Cycling, Standing) based on sensor data including accelerometer, gyroscope, magnetometer readings, and heart rate. The exploratory data analysis (EDA) revealed insights into data distributions, correlations between features, and activity patterns across different hours of the day and days of the week. The data underwent a robust preprocessing pipeline, involving the conversion of timestamp data into time-based features (hour of day, day of week), encoding of the categorical activity labels using LabelEncoder, and scaling of numerical features using StandardScaler. Several classification models, including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, and Support Vector Machine, were trained and evaluated. Logistic Regression achieved the highest accuracy of 87.25% in initial evaluations. Hyperparameter tuning was performed on the Logistic Regression model, but the initial model demonstrated comparable performance. Finally, the trained model, along with the LabelEncoder and StandardScaler, were serialized using joblib and integrated into a Flask API for potential deployment, enabling real-time activity predictions from new sensor data.